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Effects of probiotics administration on lactose intolerance in adulthood: A meta-analysis

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ABSTRACT

This meta-analysis aimed to investigate the effect of probiotic administration on adults with lactose intolerance. Twelve studies were identified from databases such as PubMed, Cochrane Library, and Web of Knowledge based on the inclusion and exclusion criteria. The effect size was estimated using the standardized mean difference (SMD), and Cochrane's Q test was used to evaluate the statistical heterogeneity of the effect size. Moderator analysis, including meta-ANOVA and meta-regression, were performed to determine the cause of heterogeneity in the effect size using a mixed-effect model. Egger's linear regression test was conducted to evaluate publication bias. The results showed that probiotic administration alleviated the symptoms of lactose intolerance, including abdominal pain, diarrhea, and flatulence. Among them, the area under the curve (AUC) showed the greatest decrease following probiotic administration (SMD, −4.96; 95% confidence interval, −6.92 to −3.00). In the meta-ANOVA test, abdominal pain and total symptoms decreased with monostrain probiotic administration. This combination was also effective for flatulence. The dosage of probiotics or lactose was significantly associated with a reduction in the total symptom score, and the linear regression models between the dosage and SMD were found to be $Y = 2.3342 \times \text{dosage} - 25.0400$ ($R^2 = 79.68\%$) and $Y = 0.2345 \times \text{dosage} - 7.6618$ ($R^2 = 34.03\%$), respectively. Publication bias was detected for most items. However, even after effect size correction, the probiotic administration effect for all items remained valid. The administration of probiotics was effective at improving adult lactose intolerance, and it is expected that the results of this study could help improve the nutritional status of adults by increasing their consumption of milk and dairy products in the future.

Key words: lactose intolerance, probiotics, meta-analysis

INTRODUCTION

Milk is a valuable food source, containing high-quality proteins, calcium, and various trace elements. However, many people cannot break down lactose smoothly in their gastrointestinal tract due to lactose intolerance, which acts as a barrier to milk consumption among adults (Messia et al., 2007). Lactose is the only disaccharide in milk, present at approximately 4.8 to 5.2% concentrations. During normal digestion, lactose is hydrolyzed into glucose and galactose by lactase (β-D-galactosidase, EC 3.2.1.23), secreted by the brush border of the intestine, which is then used as an energy source. However, when a person cannot digest lactose, the undigested lactose passes from the small intestine to the colon, causing diarrhea, abdominal pain, flatulence, and so on (Ahn et al., 2019).

Probiotics are live micro-organisms introduced from the external environment by water or food that can promote health benefits by enhancing the intestinal microbial balance (Gatesoupe, 1999; Verschuere et al., 2000; Amoah et al., 2019; Azad et al., 2019). Probiotics have been previously defined as "live micro-organisms that are ingested to provide health benefits" or "live micro-organisms, which when consumed in adequate amounts as a part of food, confer a health benefit on the host" (FAO/WHO, 2006). Probiotics can be administered orally to achieve various health benefits, including improvement of antibiotic-associated or infectious diarrhea, antiallergic effects, improvement of intestinal health, and immune response enhancement (Kechagia et al., 2013). In general, *Streptococcus thermophilus* and *Lactobacillus delbrueckii* spp. *Bulgaricus* are known to be effective against lactose intolerance. They have higher β-galactosidase (**β-gal**) activity and are more effective than other strains. Consequently, many pro-

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biotic bacteria have been shown to metabolize lactose properly (de Vrese et al., 2001; Levri et al., 2005).

Currently, the aging of the world's population is accelerating. The growth of the elderly population worldwide will continue to surpass that of the younger people over the next 35 yr. Asia leads the world in terms of the speed of aging and the size of the older population. Asian countries, such as Japan, South Korea, Hong Kong, and Taiwan, are expected to occupy all of the top ranks of the 25 oldest countries by 2050 (He et al., 2016). According to Fralic and Griffin (2001), energy needs are decreased in the elderly population; however, the requirement for nutrients, such as proteins, minerals, and specific vitamins, does not reduce. Indeed, the need for some vitamins and minerals in adults increases after 50 yr of age. Several researchers have recommended milk and dairy products as good sources of protein and calcium for aging adults. Hence, milk or dairy products can be a good source of nutrition for the elderly, in addition to the issue of adult lactose intolerance.

The ultrafiltration/reverse osmosis (UF/RO) system (Lange, 2005) has been commercialized among the many technologies used to solve the problem of lactose intolerance. This system can effectively improve lactose intolerance by selectively separating lactose from milk. However, it is difficult to expect the effect of enhancing milk calcium absorption by lactose. Therefore, this meta-analysis was performed to investigate the effect of probiotic administration on lactose intolerance relief. Furthermore, this study aimed to provide evidence to maximize the utilization of milk, which is an excellent nutritional source that can benefit adult health.

MATERIALS AND METHODS

Literature Search

No human or animal subjects were used, so this analysis did not require approval by an Institutional Animal Care and Use Committee or Institutional Review Board.

A literature search was performed in the online databases PubMed, Cochrane Library, and Web of Knowledge, following the Preferred Reporting Items for Systematic Reviews and Meta-analysis 2020 guidelines (Page et al., 2021). These 3 databases were accessed on May 4, 10, and 23, 2020, respectively. The search formula was as follows: "probiotics"[TIAB] AND ("lactose intolerance"[MeSH Terms] OR ("lactose intolerance"[MeSH Terms] OR ("lactose"[All Fields] AND "intolerance"[All Fields]) OR "lactose intolerance"[All Fields] OR "hypolactasia"[All Fields]) OR ("lactose intolerance"[MeSH Terms] OR ("lactose"[All Fields] AND "intolerance"[All Fields]) OR "lactose intolerance"[All Fields] OR ("milk"[All Fields] AND "sugar"[All Fields] AND "intolerance"[All Fields]) OR ("lactose intolerance"[MeSH Terms] OR ("lactose"[All Fields] AND "intolerance"[All Fields]) OR "lactose intolerance"[All Fields] OR ("lactose"[All Fields] AND "malabsorption"[All Fields]) OR "lactose malabsorption"[All Fields]) for PubMed ("lactose intolerance" AND probiotic*): ti, ab, kw for the Cochrane library, and probiotics [TIAB] AND "lactose intolerance" for the Web of Knowledge.

Inclusion and Exclusion Criteria

The inclusion criteria were as follows: (1) studies on lactose intolerance in adults; (2) results were expressed as mean ± standard deviation (SD) or standard error (SE), or median with quartile; (3) lactose intolerance was compared with and without probiotic administration. The corresponding exclusion criteria were as follows: (1) research on children, (2) not a research article (posters or review articles), (3) animal studies, and (4) studies with an unknown baseline.

A literature review of the collected studies was performed independently according to the Preferred Reporting Items for Systematic Reviews and Meta-analysis 2020 guidelines (Page et al., 2021) by 2 researchers (S.I.A. and M.S.K). First, articles were screened by reading the titles and abstracts. Then, the articles were classified by reading their full text when necessary. Finally, disagreements were resolved by a third researcher (B.K.H.).

Risk of Bias Assessment

The risk of bias was independently assessed using the QualSyst method (Kmet et al., 2004) by 2 researchers (M.S.K. and D.G.P.). The QualSyst method consisted of 14 questionnaires for quantitative evaluation and 10 questionnaires for qualitative assessment. The question sets comprised the following criteria: (1) description of the research objective, (2) clear and adequate study design, (3) sufficient description of subject and intervention, (4) random subject allocation into groups, (5) result measurement, and (6) sufficiently detailed reporting of outcomes. Every question was answered as "yes," "partial," "no," or "N/A," corresponding to scores of 2, 1, 0, and not applicable, respectively. After the evaluation, the assessed points from every researcher were presented as the mean ± SD. Studies with a final score of >50% were judged as high quality (Phan et al., 2022).

Data Analysis

The mean and SD were extracted from all experimental groups for the statistical analysis of the sample size. The values were estimated using Engauge Digitizer Ver. 10.12 (https://github.com/markummitchell/engauge-digitizer) if the figures presented data. The changes in lactose intolerance symptoms following probiotic administration were calculated using standardized mean difference (**SMD**) analysis (Ahn et al., 2020). The summary effect was computed with the effect size analysis using fixed- or random-effect models due to various factors, such as probiotic administration period, sample size, and probiotic strain used in each study. To evaluate the statistical heterogeneity of the effect size, Cochrane's Q test was performed with high heterogeneity when the I^2 value was $\geq$75%. Subsequently, meta-ANOVA and meta-regression analyses were performed to investigate the factors affecting the heterogeneity of the effect size. Finally, a publication bias analysis was conducted to confirm the validity of the research results and to assess publication bias in individual studies. Funnel plots were drawn to visualize publication bias, and Egger's linear regression test was performed to more accurately evaluate publication bias using numerical data. Statistical analysis was conducted using the R software (version 4.1.0, R Development Core Team, 2021, http://www.r-project.org) with 'meta' and 'metafor' packages, and all hypotheses were verified within a 5% significance level.

RESULTS AND DISCUSSION

Data Set

The data collection procedure for the meta-analysis is shown in a Preferred Reporting Items for Systematic Reviews and Meta-analysis flow diagram in Figure 1 (Page et al., 2021). The literature search yielded 278 articles, which were subsequently screened to exclude duplicates and unsuitable studies, such as those with unrelated results, no baseline, no human clinical tests, inappropriate article type, not adult subject, and so on. Table 1 shows the characteristics of the data set and the information for the meta-analysis. After the exclusion of noneligible studies, 12 studies published between 1999 and 2021 were collected. Most of them investigated abdominal pain, diarrhea, and total symptom score, and 6 further measured hydrogen gas generated after ingestion of milk, such as the area under the curve (**AUC**) and hydrogen breath test. The studies used monostrain probiotics in 5 cases and multiple strains

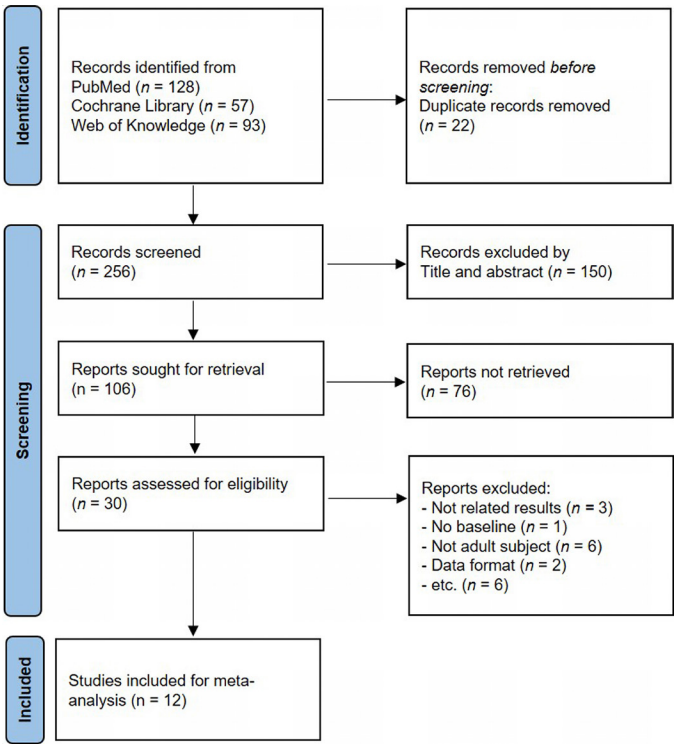


Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-analysis flow diagram showing the data collection procedure of this meta-analysis.

in 7 cases. Specifically, *Lactobacillus acidophilus* was used in 3 cases, and *Bifidobacterium* sp. was used in 6 cases. In addition, the combination of *Limosilactobacillus reuteri*, *Lacticaseibacillus casei* Shirota, *Streptococcus thermophilus*, and *L. delbrueckii* spp. *Bulgaricus* was used as a probiotic strain by Gingold-Belfer et al. (2020). A total sample size of 263 patients was included in this meta-analysis. The average amount of lactose ingested for the lactose intolerance experiment was 24.29 g, and the average probiotic administration period was 26.25 d.

Evaluation of Included Studies

Table 2 shows the QualSyst scores for the included studies. The assessment items of the QualSyst, including the study objectives, experimental design, interventions, outcome measurements, and presentation of outcomes, were described well in all studies. These studies were evaluated as high quality, with a mean of 88.525% and a SD of 0.037, with scores ranging between 96.2% and 74.8%. Thus, the included studies were judged to be of adequate quality for this meta-analysis.

Table 1. Characteristics of the data set and information collected for meta-analysis

Reference	n	Average age of subjects (yr)	Probiotics strain[1]	Probiotics dosage[2] (cfu/mL)	Probiotics administration period (d)	Lactose dosage (g)	Evaluated items[3]
Pakdaman et al., 2016	20	30–75	*Lactobacillus acidophilus*	10^{10}	28	25	Diarrhea, abdominal pain, vomiting, audible bowel sound, total symptom
Ojetti et al., 2010	20	18–65 (32 ± 5)	*Limosilactobacillus reuteri*	4×10^8	10	25	Diarrhea, abdominal pain, flatulence, bloating, total symptom
Aguilera et al., 2021	50	21–50 (28 ± 7)	*Bifidobacterium bifidum* 900791	10^5 10^7	7	20	AUC, diarrhea, abdominal pain, vomiting, audible bowel sound, nausea, total symptom
Roskar et al., 2017	19	18–75	LP+BA	5×10^9	42	24	Diarrhea, abdominal pain, vomiting, audible bowel sound, flatulence, total symptom
Jiang et al., 1996	15	24–42	*Bifidobacterium longum* B6 *B. longum* B6 (grown with glucose) *B. longum* 15708	6.23×10^8 1.02×10^9 5.84×10^8	1	20	Diarrhea, HBT, abdominal pain, audible bowel sound, flatulence, bloating
de Vrese et al., 2015	24	27.8 ± 6.2 (M) 31.5 ± 10.5 (F)	ST+LB ST+LB+lactase (from *A. oryzae*)	10^9	1	12.5	HBT, abdominal pain, flatulence, nausea
Gingold-Belfer et al., 2020	8	36.4 ± 18.6	Combination	25×10^9	168	50	Diarrhea, abdominal pain, flatulence, bloating
Almeida et al., 2012	27	23–63 (49 ± 11)	LS+BB	$(2 \times 10^8) + (5 \times 10^8)$	28	20	AUC, abdominal pain, flatulence, total symptom
Saltzman et al., 1999	10	39.7 ± 4	*L. acidophilus* BG2FO4	10^{10}	7	20	Abdominal pain, flatulence, bloating, total symptom
Pelletier et al., 2001	24	21–35 (28.4 ± 4.1)	ST+LB	$(2.2 \times 10^8) + (6.8 \times 10^8)$ $(2.7 \times 10^6) + (7.7 \times 10^6)$ $(2.6 \times 10^5) + (7.2 \times 10^5)$	1	25	AUC
Labayen et al., 2001	19	21.8 ± 1.6	ST+LB	$(1.75 \times 10^8) + (5.93 \times 10^8)$	15	25	Total symptom
Masoumi et al., 2021	27	41.4 ± 14.2	*L. acidophilus* + *Bifidobacterium* sp.	10^7	7	25	HBT

[1]M = male; F = female.

[2]LP+BA = *Lactobacillus plantarum* + *Bifidobacterium animalis*; ST+LB = *Streptococcus thermophilus* + *Lactobacillus delbrueckii* spp. *Bulgaricus*; Combination = *Lactobacillus acidophilus* + *Lacticaseibacillus rhamnosus* + *Lactobacillus casei* + *Bifidobacterium breve* + *Streptococcus thermophilus* + *Bifidobacterium longum* + *Bifidobacterium infantis*; LS+BB = *Lacticaseibacillus casei* Shirota + *Bifidobacterium breve*.

[3]AUC = area under the curve; HBT = hydrogen breath test.

Table 2. The assessed results according to the QualSyst method of the included studies

Reference	Average summary score (%)	SD between reviewer 1 and 2
Pakdaman et al., 2016	83.8	0.081
Ojetti et al., 2010	95.1	0.039
Aguilera et al., 2021	93.8	0.014
Roskar et al., 2017	94.7	0.009
Jiang et al., 1996	88.2	0.026
de Vrese et al., 2015	79.6	0.088
Gingold-Belfer et al., 2020	90.2	0.010
Almeida et al., 2012	89.5	0.006
Saltzman et al., 1999	96.2	0.021
Pelletier et al., 2001	92.9	0.044
Labayen et al., 2001	83.5	0.071
Masoumi et al., 2021	74.8	0.035

Effect of Lactose Intolerance Relief

Figure 2 shows the effect size of probiotic administration on lactose intolerance in adults. All the symptoms caused by lactose intolerance decreased. Furthermore, all parameters showed high heterogeneity ($I^2 > 75\%$). Thus, the effect sizes should be judged using a random-effects model. Flatulence (Figure 2D) showed the smallest decrease (SMD, −0.46; 95% confidence interval, −1.10 to 0.19; $P < 0.01$), whereas the AUC (Figure 2G) was most decreased by probiotics administration (SMD, −4.96; 95% CI, −6.92 to −3.00; $P < 0.01$). It is thought that probiotics help hydrolyze ingested lactose in the intestinal tract, and probiotic administration after ingestion of milk or dairy foods is useful to decrease various symptoms of lactose intolerance.

In general, lactose is digested by β-gal, an enzyme from the gut-brush border. Lactose is hydrolyzed into glucose and galactose and is used as an energy source. Milk products, including probiotics, are more useful for digesting lactose in lactose maldigesters than milk. The presence of lactase in probiotics is the main reason why probiotics can ferment milk. During digestion in the gastrointestinal tract, probiotics are lysed by bile in the small intestine, releasing enzymes that can degrade lactose (Ouwehand et al., 2002). Marteau et al. (1990) also reported that probiotics transfer lactases outside their cell membranes, with increased activity in the intestine. Based on these reports, bile-sensitive probiotics can be used for the transport of lactase or other active components to the gastrointestinal tract (Marteau and Shanahan, 2003). To describe this in more detail, at first, the probiotics which reach the digestive system act as a source of β-gal in the intestinal tract (He et al., 2008) and increase the overall hydrolytic capacity and colonic fermentation (Dhama et al., 2017). In the second step, probiotics exert antagonistic effects on heterofermentative bacteria that produce gas and secrete antibiotic-like substances (Corr et al., 2009). These bacteria competitively attach to the mucous membrane and improve colonic compensation by regulating the permeability of the intestinal barrier (Bischoff et al., 2014; Belkaid and Hand, 2014).

In contrast to this report, Kwak et al. (2012) reported that lactose can act as a calcium absorption enhancer, and the calcium in milk is absorbed and transferred to bones during the digestion of lactose. Therefore, it is thought that the administration of probiotics not only relieves lactose intolerance but also promotes the absorption of milk calcium. However, the cause of high heterogeneity in all analysis items needs to be investigated. As such, a moderator analysis was performed.

Moderator Analysis

As shown in Figure 3, probiotic administration with monostrain was more effective at reducing abdominal pain (Figure 3A; SMD, −2.28, 95% CI, −0.2.27–1.78; $I^2 = 92\%$; $P < 0.01$) and total symptoms (Figure 3D; SMD, −3.29; 95% CI, −4.82–1.76; $I^2 = 93\%$; $P < 0.01$). In contrast, multiple-strain probiotic administration was shown to be more useful to alleviate diarrhea (Figures 3B; SMD, −2.09; 95% CI, −3.53 to −0.65; $I^2 = 94\%$; $P < 0.01$) and flatulence (Figures 3C; SMD, −0.69; 95% CI, −1.63–0.24; $I^2 = 87\%$; $P < 0.01$). It has been shown that *L. acidophilus* (Saltzman et al., 1999; Pakdaman et al., 2016), *Limo. reuteri* (Ojetti et al., 2010), and *Bifidobacterium bifidum* (Aguilera et al., 2021) are effective at reducing both symptoms, including abdominal pain and total symptoms, whereas *Bifidobacterium longum* (Jiang et al., 1996) is useful only for abdominal pain.

Probiotics with proven efficacy include *Bifidobacterium*, *Saccharomyces*, and *Lactobacillus* (Usai-Satta et al., 2012). The DDS-1 strain of *L. acidophilus* has been shown to reduce symptoms of lactose intolerance when used as a daily supplement (Pakdaman et al., 2016). According to Marteau (2002), *L. acidophilus* and *B. bifidum* associated with bioyogurt are stable in bile and are thus less able to relieve lactose intolerance. Similarly, *Lacticaseibacillus rhamnosus* GG is not known to ferment lactose and is ineffective at reducing lactose intolerance symptoms (Hamilton-Miller, 2004). In a study by Jiang et al. (1996), *B. longum* B6 grown with glucose showed lower diarrhea (Figure 3B) and flatulence (Figure 3C) relief effects than other probiotics. According to this article, the carbohydrate source of the growth media results in β-gal activity and lactose transport. Similarly, Han et al. (2014) studied the effects of carbon sources on the production of β-gal by

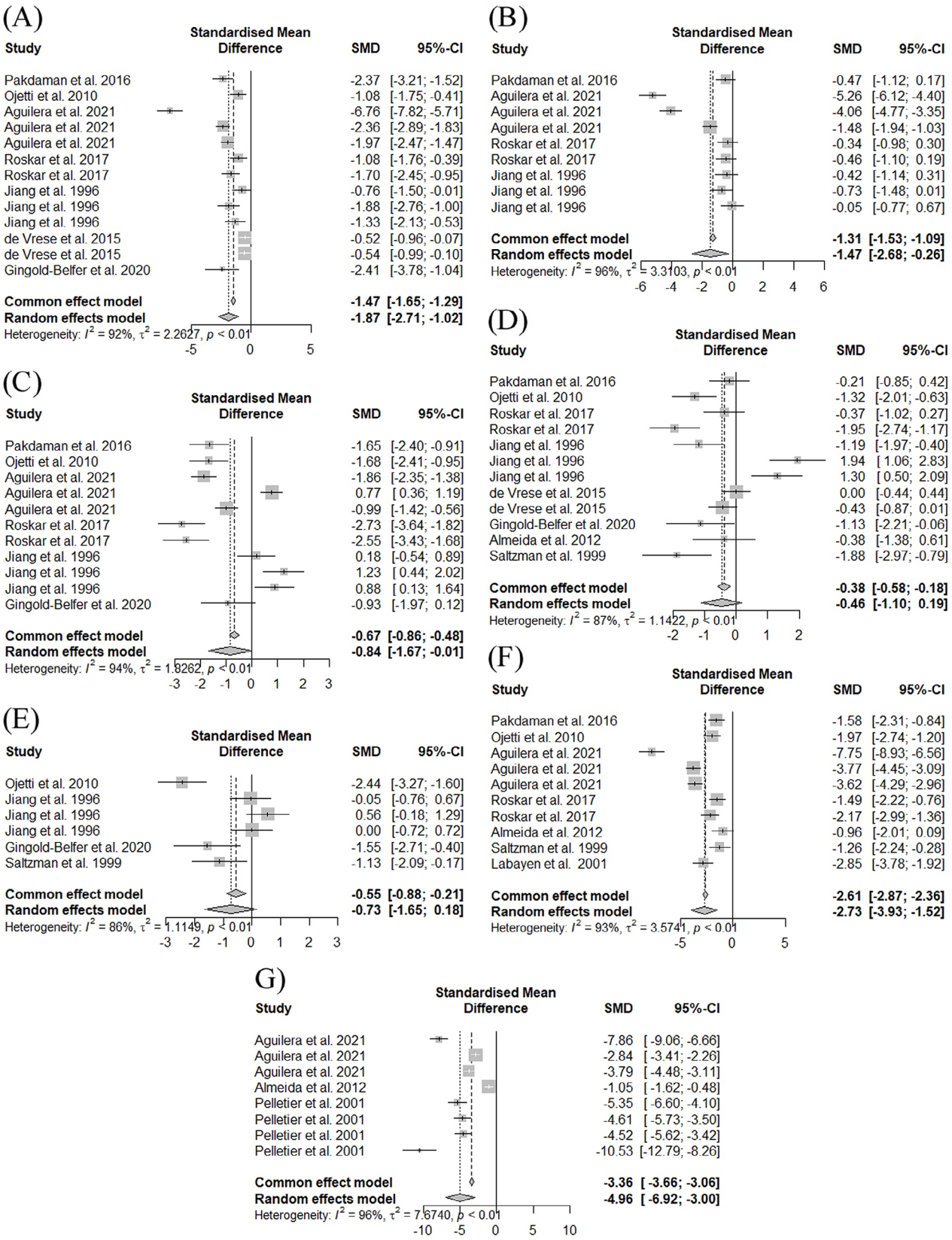


Figure 2. The effect size of probiotics on lactose intolerance in adults: (A) abdominal pain, (B) audible bowel sound, (C) diarrhea, (D) flatulence, (E) floating, (F) total symptoms, and (G) area under the curve.

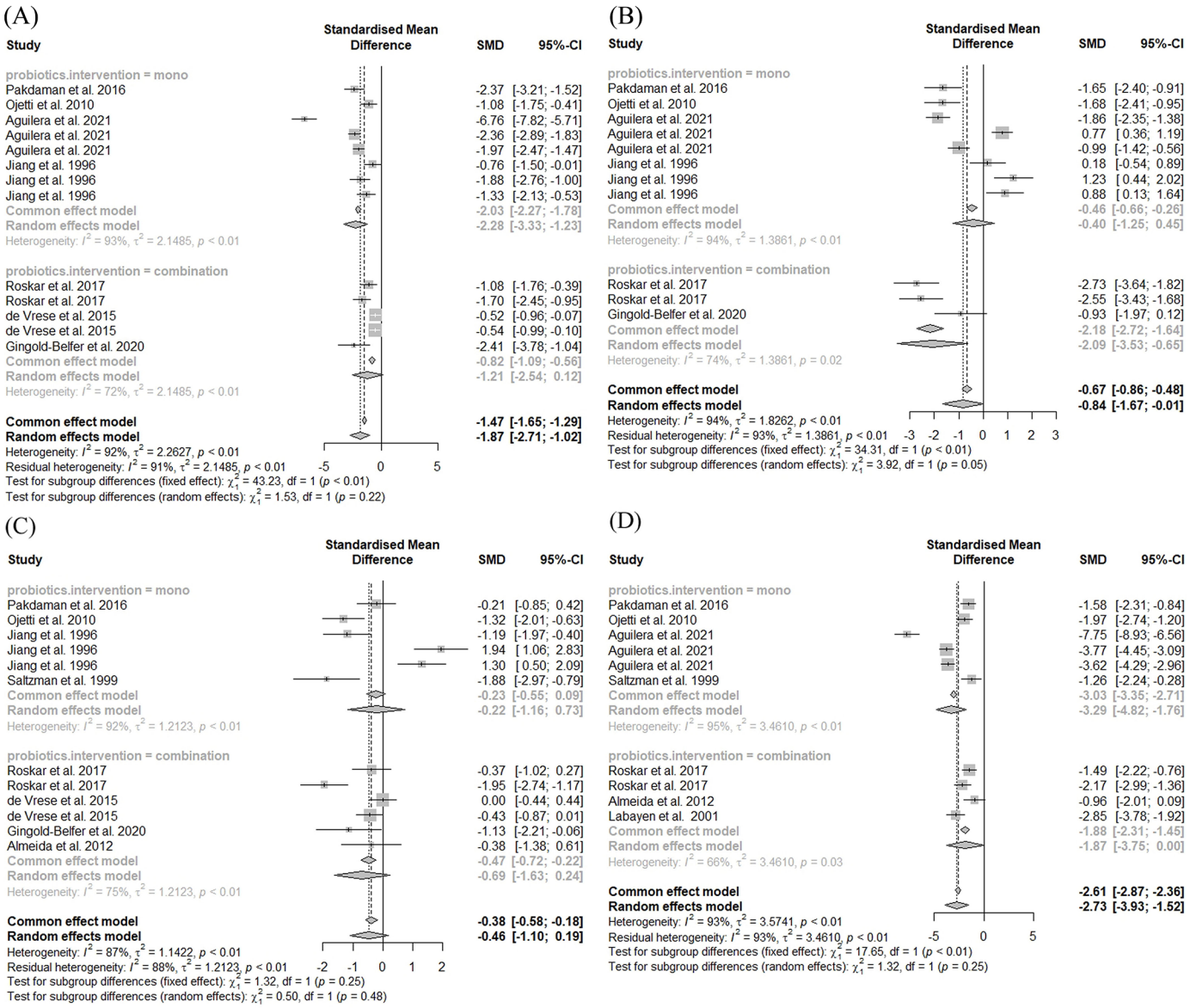


Figure 3. Subgroup analysis to analyze the effect of probiotic administration type on the adult lactose intolerance group: (A) abdominal pain, (B) diarrhea, (C) flatulence, and (D) total symptoms.

B. longum RD47. In their study, the activity of β-gal from *B. longum* RD 47 cultured with glucose was lower than that cultured with lactose or galactose. Hsu et al. (2006) and Ismail et al. (2010) reported that a 4% lactose medium could offer the best β-gal synthesis conditions.

Figure 4 shows the difference in lactose intolerance in the region by meta-ANOVA test. The region of South America showed the highest effect of abdominal pain alleviation by administering probiotics. However, Europe led the lowest alleviation effect (Figure 4A). In diarrhea and flatulence (Figures 4B and 4C), the alleviation effect in Europe was higher than in others. In contrast, North America showed the lowest reduction effect. The alleviation effect of the total symptom score (Figure 4D) of South America was the largest. However, it was shown the smallest effect size in North America. The probiotics in the human gut are related to dietary patterns. Thus, the difference in the effects of probiotics administration in each region is thought to be the difference in the intestinal microbial flora due to the difference in diet. For example, in North America and European developed countries, the westernized dietary pattern is dominant, and we detected a tendency

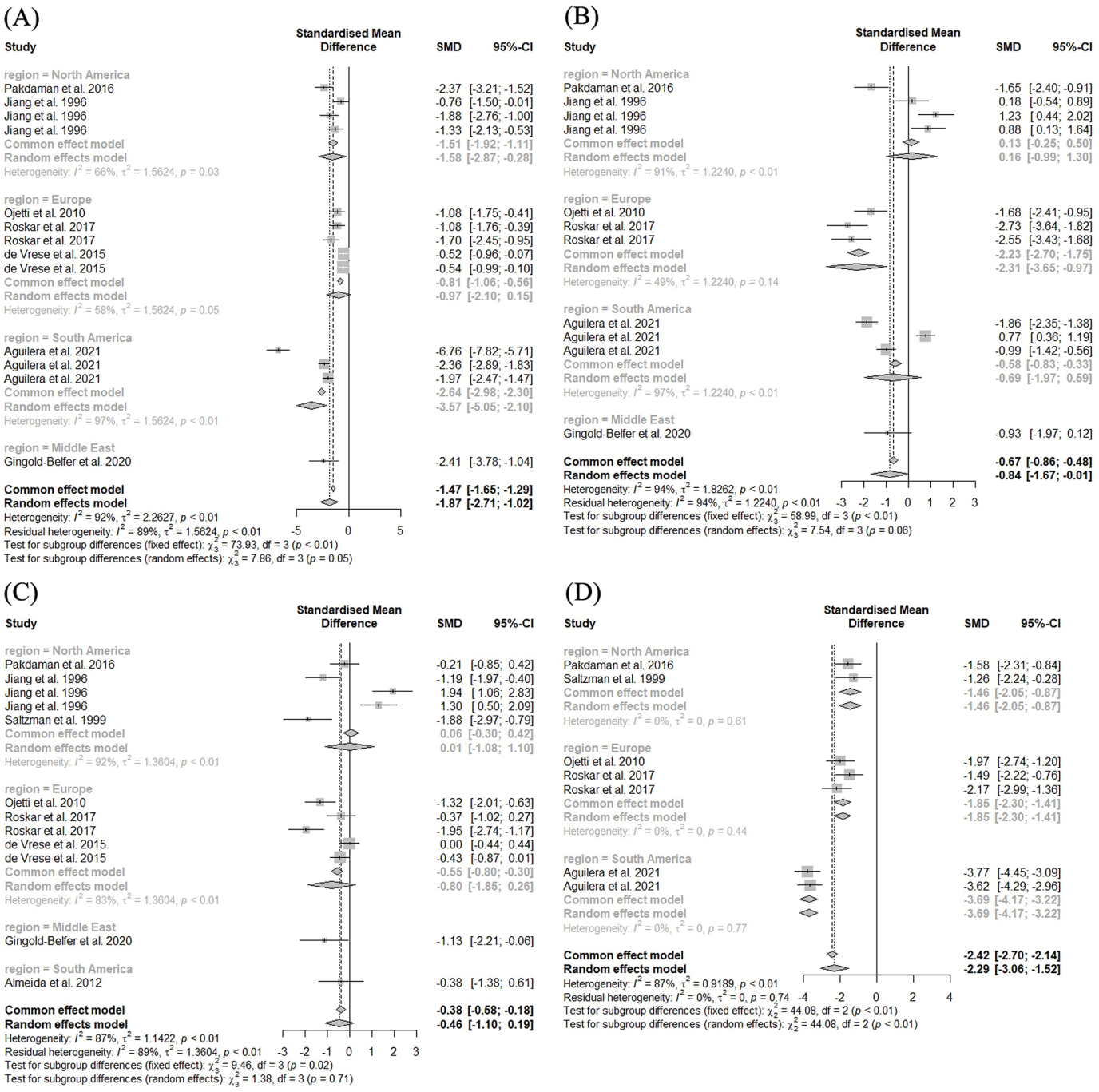


Figure 4. Subgroup analysis to analyze the effect of region on the adult lactose intolerance group: (A) abdominal pain, (B) diarrhea, (C) flatulence, and (D) total symptoms.

to consume a lot of red meat, sugar-rich beverages, or desserts, which are rich in protein, fat, and refined grains. In these cases, the genus *Bacteroides*, *Ruminococcaceae*, and *Fumicutes* tend to be found in large numbers (Chilton et al., 2015; Mobeen et al., 2018). In Asia, a starch-based diet is mainstream, and it is known that genus *Prevotella, Bacteriodes,* and *Bifidobacterium* are abundant in the intestines (Mobeen et al., 2018). Therefore, differences in regional dietary patterns cause differences in the gut microbiome, which is thought to affect the effect of reducing lactose intolerance through the administration of probiotics. This study is based on

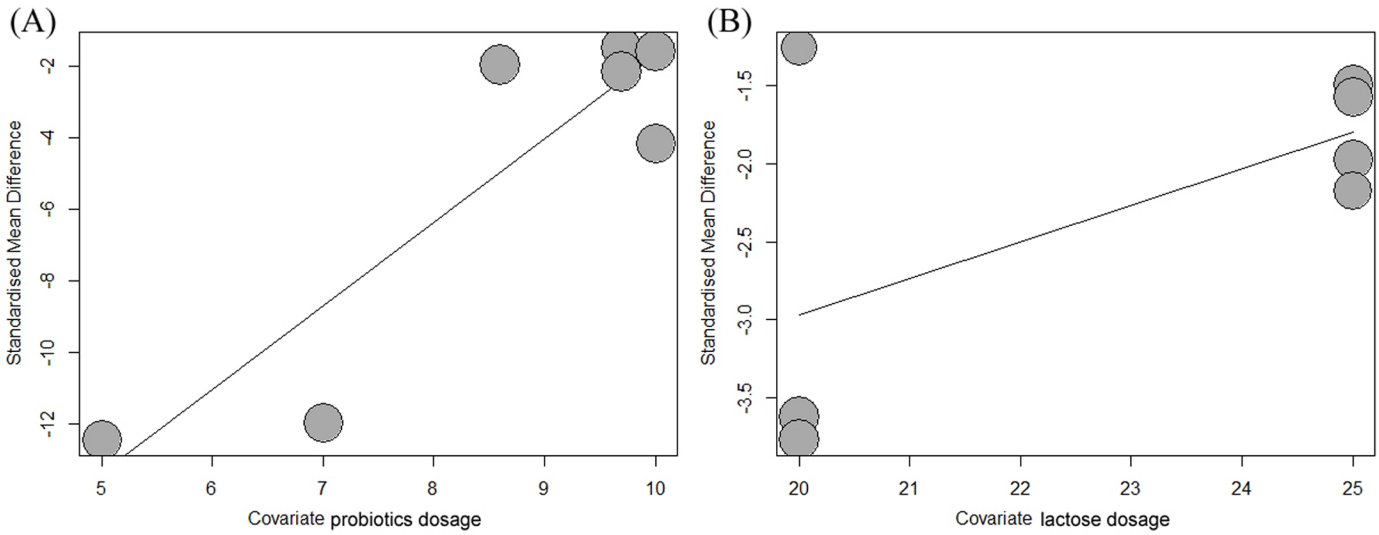


Figure 5. Bubble plot showing the total symptom reduction effect on adult lactose intolerance: (A) standardized mean difference versus probiotics dosage and (B) versus lactose dosage.

the screened 12 research articles by the inclusion/exclusion strategy and eligibility test, and it is a bit difficult to generalize and explain this phenomenon. However, it is quite interesting to note that the effect of lactose intolerance alleviation by probiotics administration varies due to differences in lifestyle, such as diet, water, or others that may differ from region to region, and further research is needed.

A meta-regression test was performed to investigate the relationship between probiotic or lactose dosage and symptoms of lactose intolerance, including abdominal pain, diarrhea, flatulence, and total symptoms. As shown in Table 3, except for the total symptoms, most items showed no significance ($P > 0.05$). Probiotic and lactose dosages were significantly associated with a reduction in the total symptom score (coefficient = 2.3342; $P < 0.0001$; $R^2 = 79.68\%$ and coefficient = 0.2345; $P = 0.0783$; $R^2 = 34.03\%$, respectively). From these results, the linear regression models between the dosage of probiotics or lactose and total symptom reduction effect (SMD) were found to be $Y = 2.3342 \times$ dosage $- 25.0400$ (95% CI:1.3546–3.3139) and $Y = 0.2345 \times$ dosage $- 7.6618$ (95% CI:13.6670–0.0266), respectively. These relationships can be expressed as

Table 3. Meta-regression test to analyze the relationship between probiotics or lactose dosage and symptoms from lactose intolerance in the adult lactose intolerance group

Item		Estimate	SE	*P*-value	CI lb[1]	CI ub[2]	R^2 (%)
Abdominal pain	Intercept	−1.1853	1.2456	0.3413	−3.6267	1.2561	0.00
	Lactose dosage	−0.0.06	0.0520	0.5559	−0.1326	0.0713	
	Intercept	−2.1955	1.1729	0.0612	−4.4945	0.1034	0.00
	Probiotics dosage	0.0890	0.1367	0.5149	−0.1790	0.3571	
Diarrhea	Intercept	0.1012	1.3578	0.9406	−2.5600	2.7624	0.00
	Lactose dosage	−0.0385	0.0529	0.4661	−0.1422	0.0651	
	Intercept	0.2585	2.5786	0.9201	−4.7955	5.3125	0.00
	Probiotics dosage	−0.0085	0.3439	0.9803	−0.6825	0.6655	
Flatulence	Intercept	0.3120	0.9025	0.7296	−1.4569	2.0810	0.00
	Lactose dosage	−0.0340	0.0371	0.3598	−0.1068	0.0388	
	Intercept	−0.3960	0.3550	0.2645	−1.0917	0.2996	0.00
	Probiotics dosage	−0.0000	0.0000	0.5573	−0.0000	0.0000	
Total symptoms	Intercept	−25.0400	4.4073	<0.0001	−33.6781	−16.4019	79.68
	Probiotics dosage	2.3342	0.4998	<0.0001	1.3546	3.3139	
	Intercept	−7.6618	3.0639	0.0124	−13.6670	−1.6566	34.03
	Lactose dosage	0.2345	0.1332	0.0783	−0.0266	0.4956	

[1]CI lb: lower limit of 95% confidence interval.

[2]CI ub: upper limit of 95% confidence interval.

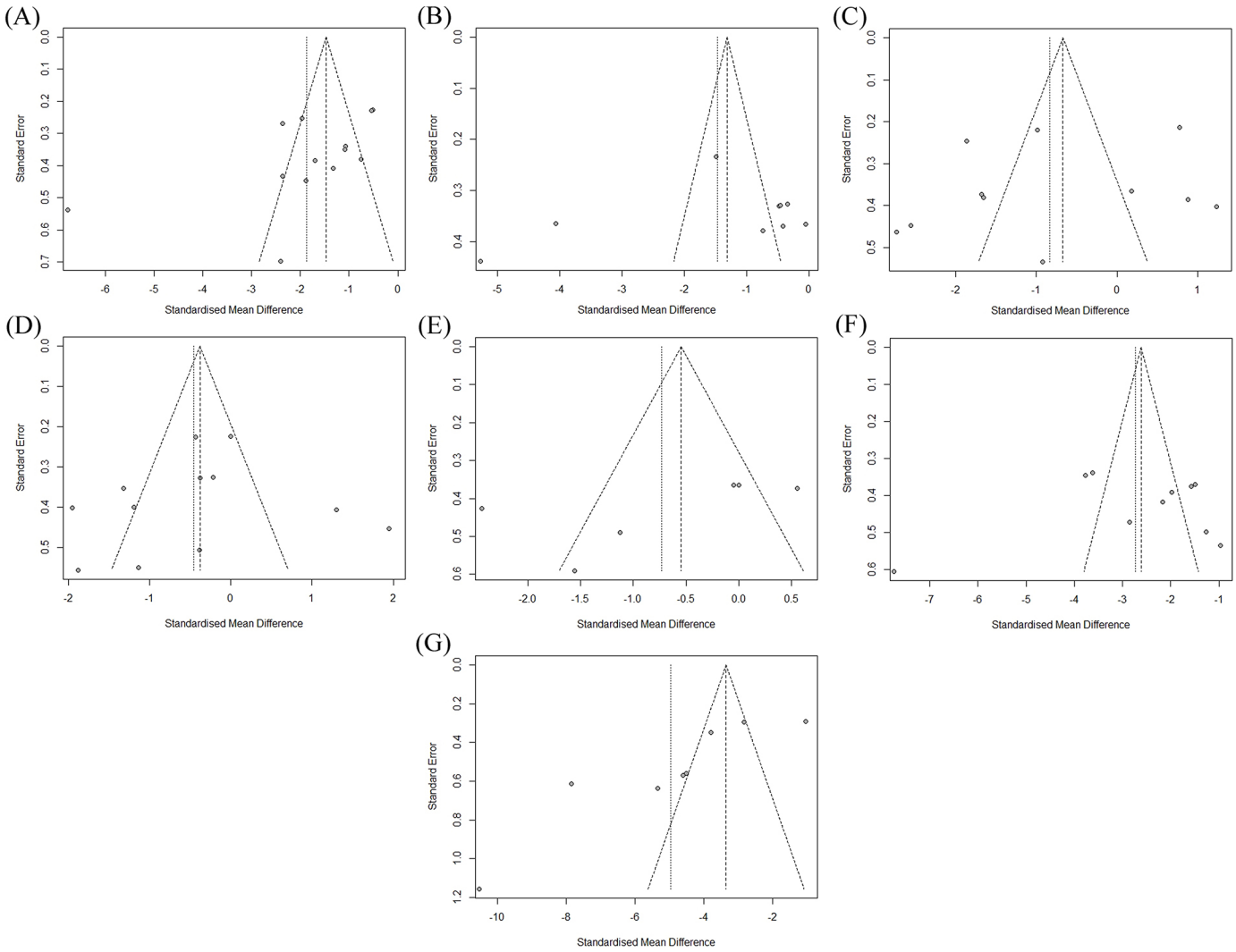


Figure 6. Funnel plots to assess publication bias: (A) abdominal pain, (B) audible bowel sound, (C) diarrhea, (D) flatulence, (E) floating, (F) total symptom, and (G) area under the curve (mg/kg × min).

bubble plots, as shown in Figure 5. As the probiotic dosage increased from 10^5 to 10^{10} cfu/mL, the SMD decreased from −12 to −2, which implies that the total symptom reduction effect decreased as the probiotic dosage increased. Similarly, an increase in lactose dosage resulted in a decrease in total symptom reduction. Appropriate probiotic administration can help improve lactose intolerance, but there are not enough studies at the exact level yet. However, the results of this study indicate that administering too many probiotics does not help much in improving lactose intolerance. Furthermore, the improvement of lactose intolerance by probiotics administration could be reduced when a large amount of lactose is consumed. Therefore, lactose dose can also act as a moderator.

Publication Bias

To investigate the presence or absence of errors in this meta-analysis, publication bias in regards to the analyzed factors, such as abdominal pain, audible bowel sounds, diarrhea, flatulence, floating total symptoms, and AUC (mg/kg × min) was assessed. Publication biases were detected for all items and corrected, as shown in Figures 6. Among these items, most dots corresponding to abdominal pain (Figure 6A) were clustered in the slightly upper part and were located inside the triangle. This suggests that studies with a relatively large number of n values were within the significance level. However, in the results of the AUC (Figure 6G), the difference between the original result (the centerline of

Table 4. Egger's linear regression test for publication bias

Item	Bias	SE bias	Intercept	SE intercept	*t*-test	df[1]	*P*-value
Abdominal pain	−6.3594	2.9904	0.5260	0.9832	−2.13	11	0.0569
Audible bowel sound	−6.8184	9.3605	0.9352	3.1288	−0.73	7	0.5826
Diarrhea	−2.8837	4.2002	0.2110	1.3469	−0.69	9	0.5097
Flatulence	−1.3287	2.7558	0.0587	0.9539	−0.48	10	0.6401
Floating	−9.4973	5.8862	3.3487	2.4464	−1.61	4	0.1819
Total symptom	−3.2078	7.1684	−1.3064	2.9700	−0.45	8	0.6664
AUC (mg/kg × min)	−10.8397	2.5261	0.9333	1.0811	−4.29	6	0.0051

[1]df: degrees of freedom of Q statistic.

the funnel) and the corrected result (external line from the funnel) seemed to be quite significant ($P < 0.05$), which indicates the presence of publication bias, meaning that correction is necessary.

Egger's linear regression test was used to confirm publication bias with statistical values, and the results are listed in Table 4. The significance between the effect size and SE was not observed for most items except AUC ($P > 0.05$). It means that there is a publication in the result of AUC. However, because some publication bias was detected in all items from the funnel plots, the results of the effect sizes were corrected using a trim-and-fill method, as indicated in Table 5. The effect size of diarrhea, abdominal pain, flatulence, and AUC were somewhat decreased, and those of audible bowel sounds and total symptoms increased. However, no significant change was observed in the effect size of diarrhea. The effect sizes of all items were significant ($P < 0.0001$).

CONCLUSIONS

This meta-analysis demonstrated the effect of probiotic administration on lactose intolerance in adults. The results of this study indicate that the administration of probiotics improved the symptoms of adult patients with lactose intolerance. In addition, based on the fact that the smooth digestion of lactose can aid in the absorption of milk calcium, we consider that a more in-depth study is needed on the relationship between the digestion of lactose and calcium absorption following the administration of probiotics. It is thought that probiotics can help lactose and calcium absorption in dairy products, and it is expected that new functionalities of probiotics will be identified through more in-depth research. In addition, further studies should aim to elucidate the relationship between the amount of lactose and probiotics added and calcium absorption in foods and dairy products for adults. The results of this study could be used to help improve the nutritional status of adults by increasing their consumption of milk and dairy products in the future.

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Table 5. The trimmed effect size of probiotics on inflammatory bowel disease in murine models

	Random effect model		Heterogeneity[2]			
Item	ES[1]	*P*-value	df	Q (*P*-value)	I^2 (%)	τ^2
Abdominal pain	−1.2919	0.0145	15	280.12 (<0.0001)	94.6	4.2814
Audible bowel sound	−1.4661	0.0176	8	181.10 (<0.0001)	95.6	3.3103
Diarrhea	−0.6426	0.1383	11	189.71 (<0.0001)	94.2	2.1087
Flatulence	−0.1920	0.5687	13	112.35 (<0.0001)	88.4	1.4170
Floating	−0.0942	0.8656	7	83.70 (<0.0001)	91.6	2.2692
Total symptom	−2.7259	<0.0001	9	129.38 (<0.0001)	93.0	3.5741
AUC (mg/kg × min)	−2.7562	0.0253	11	354.51 (<0.0001)	96.9	17.7642

[1]ES: effect size.

[2]df: degrees of freedom of Q statistic. Q: chi-squared statistic. I^2: Higgin's I^2 statistic. τ^2: true heterogeneity.

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